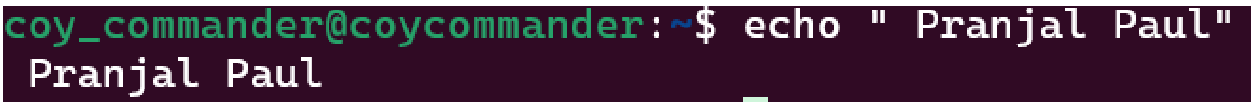
Stage 0

Project 1:**BASh Basic**

1. Print your name
echo " Pranjal Paul"



2. Create a folder titled your name
mkdir Pranjal

3. Create another new directory titled biocomputing and change to that directory with one line of command
mkdir -p biocomputing && cd biocomputing

4. Download these 3 files:
wget https://raw.githubusercontent.com/josoga2/dataset-repos/main/wildtype.fna
wget https://raw.githubusercontent.com/josoga2/dataset-repos/main/wildtype.gbk
wget https://raw.githubusercontent.com/josoga2/dataset-repos/main/wildtype.gbk

5. Move the .fna file to the folder titled your name
mv wildtype.fna ../Pranjal
6. Delete the duplicate gbk file
rm -f wildtype.gbk.1

7. Confirm if the .fna file is mutant or wild type (tatatata vs tata)

cd ../Pranjal
head wildtype.fna

#Finds exact matches of the text “tatatata” in the file wildtype.fna
grep -E "tatatata" wildtype.fna

#It Count occurrences of "tatatata" in .fna file and prints the output in a new line.
grep -o "tatatata" wildtype.fna | wc -l

```
coy_commander@coycommander:~/Pranjal$ grep -o "tatatata" wildtype.fna | wc -l
13
```

8. If mutant, print all matching lines into a new file

#since the file is mutant therefore this command is used to print the lines in a new text file.

#On using grep -E "tatatata", it shows that the file contains a mutant sequence of “tatatata”

grep -E "tatatata" wildtype.fna > mutant_lines.txt

```
coy_commander@coycommander:~/Pranjal$ grep -E "tatatata" wildtype.fna
atccacagcacctactactattactaagaacttaaaacctatataattatatataaacga
aattacttgaatattattcataatatattaacaactttattatactgctctttatatata
actatatataagtaaaacaatgatggacaatgactgtgaaaaagtatgtgatagaaacgt
aaggtatatataagattaaacattttaccttagataaagaaaatggtgcattatttaatc
gtatatatatttgccacgattgcatgtggttgtagtgcgatactgcaaggtttagcgacg
attatatatatcaattagaacattgatggttgttattacaggggaaggctcaaggtagtc
attatgcaaggtttagaagcagattcagattactaatatatataaaatttgggagtgata
gaaaaatatatataagtcattaactgtctctgcaattgttgcaacggtatcattaagtgc
aatactatatatataataccaattttaatgaaaattttaagggaggtaaataatggaaa
tagttgtcgcgcttatatatatcttcttcaaagtaattttgataagcgaattaaagatg
aatcgcccgggacgtattttaataattttcagcagttatatatataggtagaggtgttt
ctttatccataatgatagcccctatatatatctttatttacttataccctaacattatt
atctttagcggatattgaaatttttccacatcaacacgggttcaaaattatatataacag
```

9. Count number of lines (excluding header) in the .gbk file

tail -n +2 wildtype.gbk | wc -l

5749

```
coy_commander@coycommander:~/biocomputing$ tail -n +2 wildtype.gbk | wc -l
5749
```

10. Print the sequence length of the .gbk file. (Use the LOCUS tag in the first line)

echo "Sequence length:"

grep "^LOCUS" wildtype.gbk | awk '{print "Sequence Length:", $3}'

Sequence length:

Sequence Length: 197394

```
coy_commander@coycommander:~/biocomputing$ echo "Sequence length:"
 grep "^LOCUS" wildtype.gbk | awk '{print "Sequence Length:", $3}'
Sequence length:
Sequence Length: 197394
```

11. Print the source organism of the .gbk file. (Use the SOURCE tag in the first line)

echo "Source Organism: "

grep -m 1 "SOURCE" wildtype.gbk | awk '{$1=""; print "Source Organism:" $0}'

Source Organism:

Source Organism: Staphylococcus aureus

```
coy_commander@coycommander:~/biocomputing$ echo "Source Organism: "
 grep -m 1 "SOURCE" wildtype.gbk | awk '{$1=""; print "Source Organism:"   $0}'
Source Organism:
Source Organism: Staphylococcus aureus
```

12. List all the gene names of the .gbk file. Hint {grep '/gene='}

grep '/gene="' wildtype.gbk | awk -F'"' '{print "Gene:", $2}'

```
coy_commander@coycommander:~/biocomputing$ grep '/gene="' wildtype.gbk | awk -F'"' '{print "Gene:", $2}'
Gene: dnaA
Gene: dnaN
Gene: recF
Gene: gyrB
Gene: gyrA
Gene: nnrD
Gene: hutH
Gene: serS
Gene: ygaZ
Gene: metX
Gene: gdpP
Gene: rplI
```

Saving the gene list in a text file named : 'genes.txt'

```
(funtools) coy_commander@coycommander:~/biocomputing$  grep '/gene="' wildtype.gbk | awk -F'"' '{print "Gene:", $2}' >> genes.txt
```

13. Clear your terminal space and print all commands used today

clear && history

14. List the files in the two folders and share a screenshot of your terminal

Folder : Biocomputing

```
coy_commander@coycommander:~/biocomputing$ ls
wildtype.gbk
```

Folder : Pranjal

```
coy_commander@coycommander:~/Pranjal$ ls
Pranjal.sh  mutant_lines.txt  wildtype.fna
```

The file **Pranjal.sh** contains the script for this analysis

13. Clear your terminal space and print all commands used today

All the commands used today :

clear && history

14. List the files in the two folders and share a screenshot of your terminal

```
coy_commander@coycommander:~/Pranjal$ ls
Pranjal.sh  mutant_lines.txt  wildtype.fna
```

```
coy_commander@coycommander:~/biocomputing$ ls
genes.txt  wildtype.gbk
```

Project 2:**Installing Bioinformatics Software on the Terminal**

The conda environments was setup using Anaconda following these steps:

Installing of miniconda on Ubuntu

1. Download the miniconda installer

Go to the Anaconda Distribution page and copy the link for the latest Linux installer.

Then, in the terminal copy the link for linux download:

’’’’’’’’’

wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

’’’’’’’’

2. Run the installer

’’’’’’’’’

bash Miniconda3-latest-Linux-x86_64.sh

’’’’’’’’

- Type yes to accept.
- Choose the default install location (usually ~/anaconda3) unless you have a reason to change it.

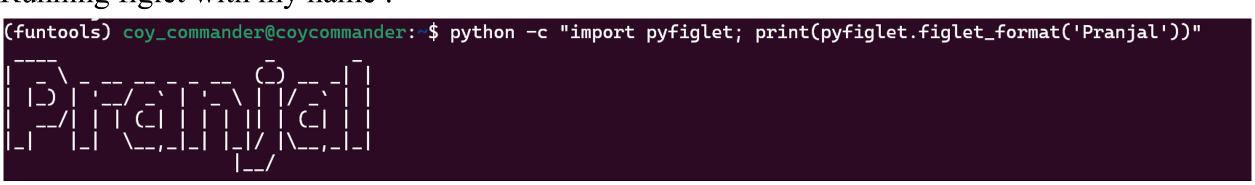
4. Initialize Conda

After installation run:

source ~/.bashrc
Then verified using :
conda --version

---

1. Activate your base conda environment
conda activate
2. Create a conda environment named funtools
conda create --name funtools python=3.10.12
3. Activate the funtools environment
conda activate funtools
4. Install Figlet using conda
conda install -c conda-forge pyfiglet
5. Run figlet <your name>
Running figlet with my name :



6 to 13 Installing tools using bioconda channel:
Configuring channels:
’’’’
conda config --add channels defaults
conda config --add channels bioconda
conda config --add channels conda-forge
’’’’
Installing multiple tools :
conda install -c bioconda bwa blast samtools bedtools spades bcftools fastp multiqc

List All Installed Packages

```
(funtools) coy_commander@coycommander:~/biocomputing$ conda list
# packages in environment at /home/coy_commander/anaconda3/envs/funtools:
#
# Name                    Version                   Build  Channel
_libgcc_mutex             0.1                        main
_openmp_mutex             5.1                       1_gnu
_python_abi3_support      1.0                  hd8ed1ab_2    conda-forge
annotated-types           0.6.0           py310h06a4308_0
attrs                     24.3.0          py310h06a4308_0
bcftools                  1.22                 h3a4d415_1    bioconda
bedtools                  2.31.1               h13024bc_3    bioconda
blas                      1.0                    openblas
blast                     2.17.0               h66d330f_0    bioconda
brotlicffi                1.0.9.2         py310h6a678d5_1
bwa                       0.7.19               h577a1d6_1    bioconda
bzip2                     1.0.8                h5eee18b_6
c-ares                    1.34.5               hef5626c_0
ca-certificates           2025.8.3             hbd8a1cb_0    conda-forge
certifi                   2025.8.3        py310h06a4308_0
cffi                      1.17.1          py310h1fdaa30_1
charset-normalizer        3.3.2              pyhd3eb1b0_0
click                     8.2.1           py310h06a4308_0
coloredlogs               15.0.1          py310h06a4308_3
colormath                 3.0.0              pyhd8ed1ab_4    conda-forge
cpython                   3.10.18         py310hd8ed1ab_0    conda-forge
curl                      8.14.1               h332b0f4_0    conda-forge
entrez-direct             24.0                 he881be0_0    bioconda
expat                     2.7.1                h6a678d5_0
fastp                     1.0.1                heae3180_0    bioconda
font-ttf-dejavu-sans-mono 2.37                 hd3eb1b0_0
font-ttf-inconsolata      2.001                hcb22688_0
font-ttf-source-code-pro  2.030                hd3eb1b0_0
font-ttf-ubuntu           0.83                 h8b1ccd4_0
fontconfig                2.14.1               hef1e5e3_0
fonts-anaconda            1                    h8fa9717_0
freetype                  2.13.3               ha770c72_1    conda-forge
gsl                       2.7.1                h6e86dc7_1
htslib                    1.22.1               h566b1c6_0    bioconda
humanfriendly             10.0            py310h06a4308_2
humanize                  4.12.2          py310h06a4308_0
icu                       70.1                 h27087fc_0    conda-forge
idna                      3.7             py310h06a4308_0
importlib-metadata        8.5.0           py310h06a4308_0
isa-l                     2.31.1               hb9d3cd8_1    conda-forge
jinja2                    3.1.6           py310h06a4308_0
jsonschema                4.25.0          py310h06a4308_0
jsonschema-specifications 2023.7.1        py310h06a4308_0
kaleido-core              0.2.1                h7c8854e_0
krb5                      1.21.3               h723845a_4
lcms2                     2.17                 h717163a_0    conda-forge
ld_impl_linux-64          2.40                 h12ee557_0
lerc                      4.0.0                h6a678d5_0
libcurl                   8.14.1               h332b0f4_0    conda-forge
libdeflate                1.24                 h86f0d12_0    conda-forge
libedit                   3.1.20230828         h5eee18b_0
libev                     4.33                 h7f8727e_1
libexpat                  2.7.1                hecca717_0    conda-forge
libffi                    3.4.4                h6a678d5_1
libfreetype               2.13.3               ha770c72_1    conda-forge
libfreetype6              2.13.3               h48d6fc4_1    conda-forge
libgcc                    15.1.0               h767d61c_4    conda-forge
libgcc-ng                 15.1.0               h69a702a_4    conda-forge
libgfortran-ng            11.2.0               h00389a5_1
libgfortran5              11.2.0               h1234567_1
libgomp                   15.1.0               h767d61c_4    conda-forge
libiconv                  1.16                 h5eee18b_3
libidn2                   2.3.4                h5eee18b_0
libjpeg-turbo             3.1.1                hb25bd0a_0
libkrb5                   1.21.3               h520c7b4_4
liblzma                   5.8.1                hb9d3cd8_2    conda-forge
liblzma-devel             5.8.1                hb9d3cd8_2    conda-forge
libnghttp2                1.64.0               h161d5f1_0    conda-forge
libnsl                    2.0.1                hb9d3cd8_1    conda-forge
libopenblas               0.3.29               ha39b09d_0
libpng                    1.6.50               h421ea60_1    conda-forge
libsqlite                 3.50.4               h0c1763c_0    conda-forge
libssh2                   1.11.1               hcf80075_0    conda-forge
libstdcxx                 15.1.0               h8f9b012_4    conda-forge
libstdcxx-ng              11.2.0               h1234567_1
libtiff                   4.7.0                h8261f1e_6    conda-forge
libunistring              0.9.10               h27cfd23_0
libuuid                   2.38.1               h0b41bf4_0    conda-forge
libwebp-base              1.6.0                hd42ef1d_0    conda-forge
libxcb                    1.17.0               h9b100fa_0
libxcrypt                 4.4.36               hd590300_1    conda-forge
libxml2                   2.9.14               h22db469_4    conda-forge
libzlib                   1.3.1                hb9d3cd8_2    conda-forge
lmdb                      0.9.31               hb25bd0a_0
markdown                  3.8             py310h06a4308_0
markdown-it-py            2.2.0           py310h06a4308_1
```